

A Tendler-Style Proof of Extension Across Takeuchi's Boundary

Standalone note

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Abstract

This note isolates a short proof that a rank-one local system which is tame along the boundary of Takeuchi's compactified Abel–Jacobi morphism extends across that boundary. The proof adapts the power argument in Tendler's treatment of tame geometric class field theory: a prime-to-the-characteristic power kills tame inertia, while the abelian fundamental group of affine space in positive characteristic is pro- p . The argument applies to a nonreduced modulus only after the ramification calculation in Chapter 2 has made the symmetric-power local system tame at the full boundary.

1 Setting

Let k be a perfect field, let C be a projective smooth geometrically connected curve over k , let $\mathfrak{m} > 0$ be an effective divisor, and put $U = C \setminus |\mathfrak{m}|$. Let Λ be a finite ring whose cardinality is invertible in k , and let $\mathcal{F}$ be a locally constant sheaf of free rank-one Λ -modules on U .

For sufficiently large d , write

$$\Phi_d: U^{(d)} \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}^d$$

for the Abel–Jacobi morphism and

$$\tilde{\Phi}_d: \tilde{C}_{\mathfrak{m}}^{(d)} \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}^d$$

for Takeuchi's compactification. It is a projective-space bundle. If

$$H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)},$$

then, over a geometric generic point of $\mathrm{Pic}_{C,\mathfrak{m}}^d$, the pair consisting of the compactified fiber and its boundary is

$$(\mathbb{P}^r, \mathbb{P}^{r-1}), \quad r = d - \deg(\mathfrak{m}) - g + 1, \tag{1}$$

and its open fiber is $\mathbb{A}^r$.

The input supplied by the ramification calculation is:

$$\mathcal{F}^{[d]} \text{ is tamely ramified at the generic point of } H. \tag{2}$$

2 The extension argument

Proposition 1. *Assume (2). Then $\mathcal{F}^{[d]}$ extends uniquely to a locally constant sheaf of free rank-one Λ -modules on $\tilde{C}_m^{(d)}$.*

Proof. Let ξ be the generic point of $\text{Pic}_{C,m}^d$ and set $K = \overline{k(\xi)}$. By (1), the geometric generic open fiber is

$$V = \mathbb{A}_K^r \subset \overline{V} = \mathbb{P}_K^r, \quad D_\infty = \overline{V} \setminus V = \mathbb{P}_K^{r-1}.$$

Let $\mathcal{P}^{[d]} = \text{Fr}(\mathcal{F}^{[d]})$ be the $\Lambda^\times$ -torsor of frames. Its restriction to V corresponds to a continuous character

$$\chi: \pi_1(V) \longrightarrow \Lambda^\times.$$

Because $\Lambda^\times$ is finite, the image of χ is finite.

Suppose first that $\text{char } k = 0$. Then $\pi_1(\mathbb{A}_K^r) = 1$; see [1, Example 4.14]. Hence χ is trivial.

Now suppose that $\text{char } k = p > 0$. Let I_∞ denote the inertia group at the generic point of D_∞ . By the tameness assumption, the finite group $\chi(I_\infty)$ has order prime to p . Let N be its exponent. Then

$$\gcd(N, p) = 1 \quad \text{and} \quad \chi^N|_{I_\infty} = 1.$$

Thus the rank-one local system corresponding to χ^N is unramified at the generic point of D_∞ . Since $\mathbb{P}_K^r$ is regular and D_∞ is the only codimension-one component of its complement, purity of the branch locus extends this local system from V to $\mathbb{P}_K^r$. Projective space is simply connected, so the extension is constant. Hence

$$\chi^N = 1 \quad \text{on } \pi_1(V). \tag{3}$$

Since χ is rank one, it factors through $\pi_1(V)^{\text{ab}}$. Tendler proves that

$$\pi_1(\mathbb{A}_K^r)^{\text{ab}} \quad \text{is a pro-}p \text{ group}$$

in [1, Example 4.15]. Consequently, the finite group $\text{im}(\chi)$ is a p -group. On the other hand, (3) says that its exponent divides N , which is prime to p . The only such p -group is the trivial group. Therefore $\chi = 1$.

We have shown in either characteristic that $\mathcal{P}^{[d]}$ is constant on the geometric generic affine fiber. In particular, its inertia at the generic point of the hyperplane at infinity is trivial. Unramifiedness can be checked after the separable extension $k(\xi) \subset K$, so $\mathcal{P}^{[d]}$ is unramified at the generic point of H .

Finally, $\tilde{C}_m^{(d)}$ is regular and H is the only codimension-one component of its complement $U^{(d)}$. Purity of the branch locus therefore extends $\mathcal{P}^{[d]}$ to a finite etale $\Lambda^\times$ -torsor on $\tilde{C}_m^{(d)}$. The associated rank-one Λ -module is the required extension of $\mathcal{F}^{[d]}$. Uniqueness follows because a dense open immersion into a normal connected scheme induces a surjection on etale fundamental groups. $\square$

3 Relation with Tendler's proof

In Tendler's tame argument, a character ρ is replaced by a power ρ^{q-1} which is unramified. The unramified geometric class field theory theorem makes the induced power of the symmetric-product character trivial on every affine Abel–Jacobi fiber. Tendler then uses the fact that

$\pi_1(\mathbb{A}^r)^{\text{ab}}$ is pro- p and that $q - 1$ is prime to p to deduce that the original character is already trivial on the fiber; see [1, §9.3].

The argument above makes two adjustments.

1. The base field here is an arbitrary perfect field, so there is no distinguished integer $q - 1$. Instead, N is chosen to be the exponent of the actual finite tame-inertia image. Finiteness of Λ guarantees that such an N exists, and tameness guarantees that N is prime to p .
2. The original local system $\mathcal{F}$ may have wild ramification allowed by the nonreduced modulus $\mathfrak{m}$. Therefore no prime-to- p power of $\mathcal{F}$ need be unramified on C . The power argument is applied only after passing to $\mathcal{F}^{[d]}$ and to the full boundary H , where the Chapter 2 calculation has already proved tameness.

Thus Tendler’s method does not replace the ramification calculation. It replaces the final step from “tame at the full boundary” to “unramified, hence extendable across the full boundary.”

Remark 1. *For rank-one local systems, this power argument proves exactly the special case needed here of the statement that affine space has trivial tame fundamental group relative to the hyperplane at infinity. It therefore provides a more explicit alternative to citing that statement directly.*

References

- [1] Avichai Tendler, *Geometric Class Field Theory*, Master’s thesis, Hebrew University of Jerusalem, 2010; arXiv:1507.00104 (2015). <https://arxiv.org/abs/1507.00104>.
- [2] Alexander Grothendieck, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Mathematics 224, Springer, 1971.